

SahuAI-PRO-Lite Beta Bug Audit Report

Repository: Saksham6062/SahuAI-PRO-Lite

File audited: Beta/index.html

Scope: Static code/configuration audit of the current Beta build.

Executive Summary

The Beta build contains several concrete bugs and configuration problems. The JavaScript has no syntax errors, so the main issues are runtime handlers, missing assets, stale UI/state references, provider configuration, and exposed credentials.

Priority Summary

Priority	Issue	Impact
Critical	Exposed API credentials	Security / quota abuse
Critical	<code>toggleResearchStage()</code> missing	Deep Research UI errors
Critical	Z.ai endpoint typo	Provider requests fail
High	Ollama Devstral model ID malformed	Model selection/request failure
High	Missing PWA/APK assets	PWA/install/download features broken
High	Hidden feature settings still referenced	Settings/state inconsistencies
Medium	Version mismatch	Debugging/release confusion
Medium	Hidden settings can reset on save	State corruption
Medium	<code>max_tokens</code> not model/provider-aware	Request failures
Low	Dark-mode placeholder contrast	Accessibility

1. Missing `toggleResearchStage()`

Deep Research markup generates calls to:

```
toggleResearchStage(...)
```

but there is no corresponding function definition.

Effect

Clicking a Deep Research stage/phase header can produce:

ReferenceError: toggleResearchStage is not defined

Suggested fix

```
function toggleResearchStage(stageId) {
  const content = document.getElementById('content-' + stageId);
  if (!content) return;

  const collapsed = content.style.display !== 'none';
  content.style.display = collapsed ? 'none' : 'block';

  const stage = document.getElementById(stageId);
  if (stage) stage.classList.toggle('collapsed', collapsed);
}
```

2. Exposed API Credentials

The frontend contains hardcoded provider credentials, including NVIDIA and TinyFish credentials.

Why this is dangerous

Anything embedded in public client-side HTML/JavaScript can be extracted by anyone.

Potential consequences:

- Unauthorized API usage
- Quota exhaustion
- Unexpected billing
- Credential abuse
- Need to rotate compromised keys

Required action

1. Revoke/rotate the exposed keys immediately.
 2. Remove them from the repository.
 3. Do not put replacement secrets in `index.html`.
 4. Put privileged API calls behind a server/worker.
 5. Use frontend-safe credentials only where the provider explicitly supports them.
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3. Z.ai Endpoint Typo

The provider configuration contains:

`https://api.z.ai/api/pass/v4/chat/completions`

The expected path is:

`https://api.z.ai/api/paas/v4/chat/completions`

Effect

Requests to the current endpoint can return 404/fail before reaching the intended API.

4. Malformed Ollama Devstral Model ID

Current configuration contains:

`devstral-small-2::24b-cloud`

There are two colons.

Expected model identifier:

`devstral-small-2:24b-cloud`

Effect

Ollama may report that the model does not exist or fail to resolve it.

5. Missing Beta Assets

The `Beta/` directory currently does not contain the local assets referenced by the HTML.

Referenced functionality includes:

- `manifest.json`
- PWA icons
- Android APK download asset
- `favicon/touch-icon` resources

Effects

- 404 requests
- Broken favicon
- PWA installation unavailable/broken
- APK download/install card broken

Fix

Either:

1. Add the referenced files to **Beta/**, or
 2. Change the paths to valid hosted assets.
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6. Five Feature Controls Are Missing

JavaScript still references feature controls such as:

```
neuralConsensusCheck  
debateModeCheck  
adaptiveReasoningCheck  
agentSwarmCheck  
constitutionalSafetyCheck
```

but those controls are absent from the current HTML.

Effect

The UI cannot enable/configure these features even though related code remains.

Some settings may silently resolve to **false**.

Fix

Choose one:

- Restore the missing UI controls, or
 - Remove the obsolete JavaScript/state handling.
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7. Version Mismatch

The UI contains v7.8 branding while source code contains v7.9.x / v7.9.12 references.

Effect

Users and developers cannot reliably identify the running build.

This complicates:

- Bug reports
- Debugging
- Release tracking
- Cache/version diagnosis

Recommended architecture

```
const APP_VERSION = '7.9.12';
```

Use that single value everywhere.

8. Hidden Settings Can Be Reset

Several settings are read using patterns equivalent to:

```
document.getElementById('someSetting')?.checked ?? false
```

If the UI control has been removed, saving settings can overwrite the existing state with the fallback value.

Effect

A user can open/save settings and unintentionally change settings that are no longer represented in the UI.

Fix

Only update state when the corresponding control actually exists:

```
const el = document.getElementById('someSetting');

if (el) {
  state.features.someSetting = el.checked;
}
```

9. max_tokens Is Not Provider/Model Aware

The application can request very large output limits, potentially up to:

64,000 tokens

The same general setting can then be passed to different providers/models.

Effect

Some providers/models may reject the request because the requested output limit exceeds their capabilities.

Fix

Store model-specific limits:

```
const maxTokens = Math.min(  
    requestedMaxTokens,  
    modelConfig.maxOutputTokens  
);
```

Also account for provider-specific API constraints.

10. Dark Mode Placeholder Contrast

Dark mode uses a relatively dim muted-text color for placeholders.

Current value is approximately:

```
--text-muted: #5d6879;
```

This can fall below the recommended WCAG AA contrast target for normal text.

Suggested improvement

Use a lighter muted color in dark mode, for example:

```
.dark-mode {  
    --text-muted: #7c8899;  
}
```

Recommended Fix Order

Immediate

1. Rotate exposed API keys.
2. Remove all secrets from the public frontend.
3. Add `toggleResearchStage()`.
4. Correct the Z.ai endpoint.
5. Correct the Devstral Ollama model ID.

Next

6. Restore or remove missing PWA/APK assets.
7. Restore/remove the five missing feature controls.
8. Prevent nonexistent settings controls from overwriting state.
9. Add provider/model-specific token limits.
10. Centralize application versioning.

Polish

11. Improve dark-mode placeholder contrast.
 12. Remove obsolete/dead settings code.
 13. Add automated static checks to CI.
 14. Add browser smoke tests for major UI workflows.
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Validation Performed

The audit included:

- HTML/JavaScript source inspection
- JavaScript syntax validation
- DOM ID/reference analysis
- Function-definition checks
- Provider endpoint inspection
- Model identifier inspection
- Local asset reference inspection
- Storage/state handling inspection
- Existing Beta bug report review

Important Result

The inline JavaScript **parses successfully**. The primary problems are therefore not general syntax corruption but concrete runtime/configuration/security issues.

Final Assessment

The Beta build is structurally functional but should not be considered release-ready until the security and provider configuration issues are fixed.

Highest-impact fixes

Security: remove/rotate exposed credentials

Runtime: implement `toggleResearchStage()`

Provider: fix Z.ai endpoint

Model: fix Devstral model ID

Distribution: restore missing PWA/APK assets

State: clean up removed feature controls and settings persistence